

Integration by Parts Worksheet — Worked Answer Key

Twenty problems covering polynomial products, logarithms, inverse trigonometric functions, definite integrals, and cyclic cases.

Revision: 2026-07-23 **Canonical page:** <https://bettergrades.net/subjects/math/calculus/worksheets/integration-by-parts/>

1. Evaluate $\int x e^x dx$.

Answer: $e^x(x - 1) + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- Simplification gives $e^x(x - 1) + C$.

2. Evaluate $\int x^2 e^x dx$.

Answer: $e^x(x^2 - 2x + 2) + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- Simplification gives $e^x(x^2 - 2x + 2) + C$.

3. Evaluate $\int x^3 e^x dx$.

Answer: $e^x(x^3 - 3x^2 + 6x - 6) + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- Simplification gives $e^x(x^3 - 3x^2 + 6x - 6) + C$.

4. Evaluate $\int x e^{2x} dx$.

Answer: $e^{2x}(2x - 1)/4 + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- Simplification gives $e^{2x}(2x - 1)/4 + C$.

5. Evaluate $\int x \sin x dx$.

Answer: $-x \cos x + \sin x + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- Simplification gives $-x \cos x + \sin x + C$.

6. Evaluate $\int x \cos x dx$.

Answer: $x \sin x + \cos x + C$

- Choose u as the factor that simplifies when differentiated and integrate dv .
- Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $x \sin x + \cos x + C$.

7. Evaluate $\int x^2 \sin x \, dx$.

Answer: $-x^2 \cos x + 2x \sin x + 2 \cos x + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $-x^2 \cos x + 2x \sin x + 2 \cos x + C$.

8. Evaluate $\int x^2 \cos x \, dx$.

Answer: $x^2 \sin x + 2x \cos x - 2 \sin x + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $x^2 \sin x + 2x \cos x - 2 \sin x + C$.

9. Evaluate $\int \ln x \, dx$.

Answer: $x \ln x - x + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $x \ln x - x + C$.

10. Evaluate $\int x \ln x \, dx$.

Answer: $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $\frac{x^2}{2} \ln x - \frac{x^2}{4} + C$.

11. Evaluate $\int (\ln x)^2 \, dx$.

Answer: $x[(\ln x)^2 - 2 \ln x + 2] + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $x[(\ln x)^2 - 2 \ln x + 2] + C$.

12. Evaluate $\int \arctan x \, dx$.

Answer: $x \arctan x - \frac{1}{2} \ln(1 + x^2) + C$

a. Choose u as the factor that simplifies when differentiated and integrate dv .

b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.

c. Simplification gives $x \arctan x - \frac{1}{2} \ln(1 + x^2) + C$.

13. Evaluate $\int \arcsin x \, dx$.

Answer: $x \arcsin x + \sqrt{1 - x^2} + C$

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification gives $x \arcsin x + \sqrt{1-x^2} + C$.

14. Evaluate $\int_0^1 x e^x dx$.

Answer: 1

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives 1.

15. Evaluate $\int_0^\pi x \sin x dx$.

Answer: π

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives π .

16. Evaluate $\int e^x \cos x dx$.

Answer: $\frac{e^x}{2}(\sin x + \cos x) + C$

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives $\frac{e^x}{2}(\sin x + \cos x) + C$.

17. Evaluate $\int e^x \sin x dx$.

Answer: $\frac{e^x}{2}(\sin x - \cos x) + C$

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives $\frac{e^x}{2}(\sin x - \cos x) + C$.

18. Evaluate $\int x^4 e^x dx$.

Answer: $e^x(x^4 - 4x^3 + 12x^2 - 24x + 24) + C$

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives $e^x(x^4 - 4x^3 + 12x^2 - 24x + 24) + C$.

19. Evaluate $\int x^2 \ln x dx$.

Answer: $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u dv = uv - \int v du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives $\frac{x^3}{3} \ln x - \frac{x^3}{9} + C$.

20. Evaluate $\int_1^e \ln x \, dx$.

Answer: 1

- a. Choose u as the factor that simplifies when differentiated and integrate dv .
- b. Apply $\int u \, dv = uv - \int v \, du$; repeat when the remaining integral still has a polynomial factor.
- c. Simplification and endpoint evaluation gives 1.